

Total No. of Questions : 4]

P-5326

SEAT No. :

[Total No. of Pages : 1

[6188]-295

B.E. (Artificial Intelligence and Data Science Engineering) (Insem.)
BIOINFORMATICS
(2019 Pattern) (Semester - VII) (417523(D)) (Elective - III)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) What is Bioinformatics? Give its applications in various fields. [5]
 b) How can molecular biology be considered as an information science. [5]
 c) Explain the function of DNA, RNA & protein. [5]

OR

- Q2)** a) Explain the function of biological molecules. [5]
 b) Discuss the importance of cellular process. [5]
 c) What is the difference between structural & functional genomics. [5]
- Q3)** a) Explain RNA sequencing & analysis. [5]
 b) Explain DNA sequence analysis. [5]
 c) Explain NCBI dataset & repositories. [5]

OR

- Q4)** a) Define Intron, Exon & Microarray. [5]
 b) Explain in detail with RNA sequence Genome annotation. [5]
 c) Discuss the importance of gene prediction. [5]

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